

ECONOMETRICS ASSIGNMENT I



ANDHRA PRADESH

Estimation of the Energy (or Calories), Protein, and Fat Demand
Elasticity for Rural Areas

Submitted to

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1 Estimate the protein demand model using the following dependent (first variable) and independent variables (the remaining variables) using sampling weights (per_wgt=hh_size*weights). Summarise your findings using R2 or Adjusted R2 (wherever applicable), and the interpretation of the slope coefficient based on marginal effect and elasticity.

(a) ProteinHH (Y), mpce (X)

Dependent Variable(Y): ProteinHH

Independent Variable(X): mpce

Estimation Method: Weighted Least Squares

Model estimated is:

$$ProteinHH = \beta_0 + \beta_1 mpce + u_i$$

Estimated using sampling weights defined as:

$$Per_wgt = hh_size \times weights$$

Table 1.1: Coefficients of Model 1

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	310.585020	8.815552	35.23	< 2e-16 ***
mpce	-0.010850	0.001573	-6.90	7.02e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Estimated Regression Equation:

$$ProteinHH = 310.5850 - 0.01085mpce$$

Intercept (β_0) = 310.585020

mpce coefficient (β_1) = -0.010850

The coefficient of mpce is -0.01085

Interpretation:

Holding other factors constant,

- If mpce **increases by 1 unit**, ProteinHH **decreases by 0.01085** units on average.

Because the coefficient is **negative**, the relationship is **inverse**.

Goodness of fit

$R^2 = 0.02382$

Interpretation:

This means, Only **2.38% of variation** in ProteinHH is explained by mpce. So, the explanatory power of the model is **very low**, even though the coefficient is **statistically significant**.

(Since this model contains **only one independent variable**, reporting **R-squared** is **appropriate**)

Elasticity

The elasticity of ProteinHH with respect to mpce is:

Since the model is in level-level form, income elasticity is calculated as:

$$E = \beta_1 \times \frac{\bar{X}}{\bar{Y}}$$

E = -0.2935082

Since β_1 is **negative and small**, ProteinHH is **relatively inelastic**.

Overall Model Significance

F-statistic = 47.61

p-value = 7.017e-12 (i.e., 0.000000000007017)

Since p-value < 0.01

We reject the null hypothesis and conclude that the model is **statistically significant at the 1% level.**

(b) Per capita ProteinHH (Y), mpce (X)

Dependent Variable(Y): Per Capita ProteinHH

Independent Variable(X): mpce

Estimation Method: Weighted Least Squares

Model estimated is:

$$\text{Per capita ProteinHH} = \beta_0 + \beta_1 \text{mpce} + u_i$$

Estimated using sampling weights defined as:

$$\text{per_wgt} = \text{hh_size} \times \text{weights}$$

Table 1.2: Coefficients of Model 2

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.513e+01	8.655e-01	40.59	<2e-16 ***
mpce	5.264e-03	1.544e-04	34.09	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Estimated Regression Equation:

$$\text{Per capita ProteinHH} = 35.13 + 0.005264\text{mpce}$$

Intercept (β_0) = $3.513\text{e}+01 = 35.13$

mpce coefficient (β_1) = $5.264\text{e}-03 = 0.005264$

The coefficient of mpce is **0.005264**

Interpretation:

Holding other factors constant,

- If mpce **increases by 1 unit**, Per capita protein consumption **increases by 0.005264** units on average.

Since the coefficient is **positive**. There is a **positive** relationship between **mpce** and **per capita protein consumption**.

Goodness of fit

$$R^2 = 0.3734$$

Interpretation:

About **37.34%** of the variation in per capita protein consumption is explained by mpce.

This is a much stronger model compared to **Part (a) (which had only 2.38%)**.

(Since this model contains **only one independent variable**, reporting **R-squared** is **appropriate**)

Elasticity

The income elasticity of Per capita ProteinHH with respect to mpce is:

Since the model is in level-level form, income elasticity is calculated as:

$$E = \beta_1 \times \frac{\bar{X}}{\bar{Y}}$$

$$E = 0.4456837$$

Per capita Protein HH is an **relatively inelastic**.

This implies that Protein consumption increases as mpce increases. But, the increase is less than proportionate.

Overall Model Significance

F-statistic = 1162

p-value < 2e-16 (i.e., less than 0.0000000000000002)

Since p-value < 0.01

We reject the null hypothesis and conclude that the model is **statistically significant at the 1% level**.

(c) ln(Per capita ProteinHH) (Y), ln_mpce (X)

Dependent Variable(Y): ln(Per Capita ProteinHH)

Independent Variable(X): ln(mpce)

Estimation Method: Weighted Least Squares

Model estimated is:

$$\ln(\text{Per capita ProteinHH}) = \beta_0 + \beta_1 \ln(\text{mpce}) + u_i$$

Table 1.3: Coefficients of Model 3

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.20463	0.11378	-1.798	0.0723
ln_mpce	0.50663	0.01335	37.955	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Estimated Regression Equation:

$$\ln(\text{Per capita ProteinHH}) = -0.20463 + 0.50663\ln(\text{mpce})$$

Intercept (β_0) = -0.20463

$\ln_mpce$ coefficient (β_1) = 0.50663

The coefficient of mpce is 0.50663

Interpretation:

Holding other factors constant,

- If mpce **increases by 1 percentage**, Per capita protein consumption **increases by 0.5264 percentage** on average.

Since the coefficient is **positive**. There is a **positive** relationship between **mpce** and **per capita protein consumption**.

Goodness of fit

$$R^2 = 0.4247$$

Interpretation:

About **42.47% of the variation** in log per capita protein consumption is explained by log mpce.

This model fits better than **Part (b) (37%)** and much better than **Part (a) (2.3%)**.

(Since this model contains **only one independent variable**, reporting **R-squared** is **appropriate**)

Elasticity

In a log-log regression, the coefficient of the independent variable is directly the elasticity.

So, Elasticity=**0.50663**

Interpretation

A **1% increase in mpce** leads to a **0.5066% increase in per capita protein consumption**, on average.

Since $0 < 0.5066 < 1$

Per Capita Protein HH consumption is **relatively inelastic**.

This means $\ln(\text{Per Capita protein HH})$ increases with $\ln(\text{mpce})$, but less than proportionately.

Overall Model Significance

F-statistic = 1441

p-value < $2.2e-16$ (i.e., less than 0.00000000000000022)

Since p-value < 0.01

We reject the null hypothesis and conclude that the model is **statistically significant at the 1% level**.

(d) $\ln(\text{Per capita ProteinHH})$ (Y), hh_size (X)

Dependent Variable (Y): $\ln(\text{Per Capita ProteinHH})$

Independent Variable (X): hh_size

Model estimated is:

$$\ln(\text{Per capita ProteinHH}) = \beta_0 + \beta_1 \text{hh_size} + u_i$$

Table 1.4: Coefficients of Model 4

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.275513	0.010435	409.71	<2e-16 ***
hh_size	-0.038583	0.002098	-18.39	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Estimated Regression Equation:

$$\ln(\text{Per capita ProteinHH}) = -4.275513 - 0.038583 \text{ hh_size}$$

Intercept (β_0) = -4.275513

ln_mpce coefficient (β_1) = 0.038583

The coefficient of mpce is -0.038583

Goodness of fit

$$R^2 = 0.1477$$

Interpretation:

About **14.77% of variation** in log per capita protein consumption is explained by household size. This model explains **less variation** compared to the **(42%)**, but **more than part(a)**.

(Since this model contains **only one independent variable**, reporting **R-squared** is **appropriate**)

Elasticity

Since this is a log-linear model (log of Y, X in levels):

The coefficient measures percentage change in Y for one unit change in X.

Exact percentage effect:

$$\% \Delta Y = (e^{\beta_1} - 1) * 100$$

$$= (e^{-0.038583} - 1) * 100$$

$$\approx (-3.79\%)$$

Alternatively,

Average elasticity uses the weighted mean (or simple mean) of X to summarize elasticity across the sample. **E = β_1 * 100**

$$E = -3.853$$

Interpretation:

So, one-member increase in household size reduces per capita protein consumption by approximately 3.8%.

Larger households consume **less protein per person**.

There is a **negative relationship**. This may reflect resource sharing within households.

Overall Model Significance

F-statistic = 338.1

p-value < 2.2e-16 (i.e., less than 0.00000000000000022)

Since p-value < 0.01

We reject the null hypothesis and conclude that the model is **statistically significant at the 1% level**.

(e) ln(Per capita ProteinHH) (Y), hh_size(X2), ln_mpce (X3)

Dependent Variable(Y): ln(Per Capita ProteinHH)

Independent Variables(X2 and X3): hh_size and ln(MPCE)

Multiple regression model:

$$\ln(\text{Per capita_proteinHH}) = \beta_0 + \beta_1 \text{hh_size} + \beta_2 \ln(\text{mpce}) + u$$

Table 1.5: Coefficients of Model 5

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.178541	0.130509	1.368	0.171
hh_size	-0.011204	0.001918	-5.841	6.07e-09 ***
ln_mpce	0.467264	0.014853	31.459	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Estimated Regression Equation

$$\ln(\text{Per capita ProteinHH}) = 0.1785 - 0.011204 \text{ hh_size} + 0.467264 \ln(\text{mpce})$$

Intercept = 0.178541

hh_size coefficient = -0.011204

ln_mpce coefficient = 0.467264

The coefficient of mpce is 0.467264

Interpretation:

A 1% increase in mpce increases per capita protein consumption by 0.467%, holding household size constant.

A one-member increase in household size reduces per capita protein consumption by approximately 1.11%, holding mpce constant.

Goodness of fit

Adjusted R² = 0.4341

Interpretation:

This means, after adjusting for the number of regressors, **43.41% of the variation** in log per capita protein consumption is explained by household size and income.

The model has **moderate explanatory power** and the **variables** included are **relevant**
The fit is better than all previous models

(Since this is a **multiple regression**, **Adjusted R² is more appropriate than simple R²** because it accounts for **extra regressors**)

Elasticity

Elasticity of hh_size (X2)

- Coefficient $\beta_1 = -0.011204$

$$E_{hh_size} = -0.011204 \times 100 = -1.1204\%$$

Alternatively,

Exact percentage effect:

$$\% \Delta Y = (e^{\beta_1} - 1) * 100$$

$$= (e^{-0.011204} - 1) * 100$$

$$\approx (-1.114\%)$$

Interpretation:

A one-member increase in household size reduces per capita protein consumption by approximately 1.12%.

Elasticity of $\ln_mpce$ (X3)

- Coefficient $\beta_2 = 0.467264$

Since MPCE is in log form, **the coefficient itself is the elasticity:**

$$E_{\ln mpce} = 0.467264$$

Interpretation:

A 1% increase in MPCE leads to approximately 0.467% increase in per capita protein consumption.

Overall Model Significance

F-statistic = 749.6

p-value < $2.2e-16$ (i.e., less than 0.00000000000000022)

Since p-value < 0.01

We reject the null hypothesis and conclude that the model is **statistically significant at the 1% level.**

2 (a) Estimate the following two additional models for energy and fat as in (e) above. Report these results for Energy, Protein and Fat together in one table and compare the elasticities.

$\ln(\text{Per capita Energy}_{HH})$ (Y), hh_size (X2), $\ln_mpce$ (X3)

$\ln(\text{Per capita Fat}_{HH})$ (Y), hh_size (X2), $\ln_mpce$ (X3)

Table 2.1: Regression Results for Energy, Protein & Fat Demand

Coefficients			
<i>Variable</i>	<i>Protein</i>	<i>Energy</i>	<i>Fat</i>
Intercept	-0.720*** (0.082)	3.501*** (0.071)	-1.140*** (0.093)
hh_size	0.001 (0.002)	-0.005*** (0.001)	-0.008*** (0.002)
ln_mpce	0.567*** (0.009)	0.491*** (0.008)	0.621*** (0.011)
Statistical Results			
<i>Num.Obs.</i>	6302	6302	6302
R^2	0.385	0.391	0.378
$R^2 Adj.$	0.385	0.391	0.378
<i>AIC</i>	3875.9	2031.7	5195.6

<i>BIC</i>	3896.1	2051.9	5215.8
<i>RMSE</i>	0.33	0.28	0.37
Std.Errors	IID	IID	IID

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Comparison of Elasticities

$$\ln(\text{per capita Nutrient}) = \beta_0 + \beta_1 \text{hh_size} + \beta_2 \ln(\text{mpce})$$

where the dependent variable is alternatively per-capita Energy, Protein, and Fat intake. The estimated results are reported jointly in one table for Energy, Protein, and Fat.

Income Elasticity (In_mpce)

The coefficient of $\ln_mpce$ represents the elasticity of demand because both dependent and independent variables are in logarithmic form.

The estimated elasticities are:

- Energy elasticity = 0.491
- Protein elasticity = 0.567
- Fat elasticity = 0.621

These results indicate that a 1% increase in monthly per capita expenditure (MPCE) increases per-capita energy intake by about 0.491%, protein intake by about 0.567%, and fat intake by about 0.621%.

Comparing elasticities, fat demand is the most responsive to income changes, followed by protein, while energy demand is the least responsive. This suggests that households increase fat and protein consumption relatively more than energy intake as income rises.

Household size effect (hh_size)

The coefficient of household size is negative for energy and fat, indicating that larger households tend to have lower per-capita intake of these nutrients, while the effect on protein is statistically insignificant. Thus, when an additional member is added to the household, the energy and fat intake reduces by 0.5% and 0.8% respectively. On the other hand, protein intake increases by 1%.

Since household size is not in logarithmic form, its precise effect is interpreted using:

$$\% \Delta Y = (e^{\beta_1} - 1) \times 100$$

The estimated effects are:

Protein ($\beta_1 = 0.001$):

$$(e^{0.001} - 1) \times 100 \approx 0.1\%$$

A one-unit increase in household size increases per capita protein consumption by approximately 0.1% (very small and statistically insignificant).

Energy ($\beta_1 = -0.005$):

$$(e^{-0.005} - 1) \times 100 \approx -0.5\%$$

A one-unit increase in household size reduces per capita energy consumption by approximately 0.5%.

Fat ($\beta_1 = -0.008$):

$$(e^{-0.008} - 1) \times 100 \approx -0.8\%$$

A one-unit increase in household size reduces per capita fat consumption by approximately 0.8%.

Overall Interpretation

The coefficient of household size is negative for energy and fat, indicating that larger households tend to have lower per-capita intake of these nutrients, suggesting **economies of scale in consumption**. The effect on protein is very small and statistically insignificant.

(b) Test if the each of the energy, protein and fat demand is inferior or normal or luxury 'good'?

The classification of goods is based on the estimated income elasticity of demand obtained from the coefficient of $\ln_mpce$ in the log–log regression model.

The interpretation rules are:

- Elasticity $< 0 \rightarrow$ Inferior good

- $0 < \text{Elasticity} < 1 \rightarrow \text{Normal (necessity) good}$
- $\text{Elasticity} > 1 \rightarrow \text{Luxury good}$

The estimated income elasticities are:

- Energy elasticity = 0.491
- Protein elasticity = 0.567
- Fat elasticity = 0.621

Since all three elasticities are **positive but less than one**, energy, protein and fat are classified as **normal necessity goods**. This indicates that increases in income lead to higher nutrient consumption, but the increase is less than proportional to income growth. None of the nutrients are inferior goods or luxury goods

3. Test which of the following two models fit the data better and interpret the coefficient of household size as in (a) and (b).

(a) $\ln(\text{Per capita ProteinHH})$ (Y), $\ln(\text{hh_size})$ (X21), $\ln(\text{mpce})$ (X3)

(b) $\ln(\text{Per capita ProteinHH})$ (Y), $\ln(\text{hh_size})$ (X22), $\ln(\text{mpce})$ (X3)

If the model fit is very similar for either of the models which one would you prefer to use for ease of interpretation and why?

a) $\ln(\text{Per capita ProteinHH})$ (Y), $\ln(\text{hh_size})$ (X21), $\ln(\text{mpce})$ (X3)

The Model (a) is specified as:

$$\ln(\text{pc_protein}) = \beta_0 + \beta_1 \ln(\text{hh_size}) + \beta_2 \ln(\text{mpce})$$

Table 3.1: Coefficients of Model (a)

	<i>Estimate</i>	<i>Std_Error</i>	<i>t_value</i>	<i>p_value</i>
<i>Intercept</i>	0.1785	0.1305	1.3680	0.1715

<i>hh_size</i>	-0.0112	0.0019	-5.8410	6.07e-09 ***
<i>ln_mpce</i>	0.4673	0.0149	31.4591	< 2e-16 ***

Model (a) regresses the logarithm of per capita protein consumption on household size in levels and the logarithm of monthly per capita consumption expenditure (MPCE). The estimated intercept is 0.1785, with a standard error of 0.1305 and a t-statistic of 1.368. Since the t-statistic is small in absolute value, the intercept is not statistically significant at conventional levels. The coefficient on household size is -0.0112 , with a standard error of 0.0019 and a t-statistic of -5.841 , which is statistically significant at the 1 per cent level. It indicates that an additional household member reduces per capita protein consumption by approximately 1.12 per cent, holding everything else constant. The coefficient on $\ln(\text{mpce})$ is 0.4673, with a standard error of 0.01485 and a t-statistic of 31.459, and is statistically significant at 1 per cent. This implies that a 1 per cent increase in MPCE increases per capita protein consumption by approximately 0.47 per cent, keeping everything else constant. The R-square of the model is 0.4346, meaning that about 43.5 per cent of the variation in per capita protein consumption is explained by the regressors. The Adjusted R-square is 0.4341, indicating a very similar explanatory power after adjusting for degrees of freedom.

b) $\ln(\text{Per capita ProteinHH})$ (Y), $\ln(\text{hh_size})$ (X22), $\ln_mpce$ (X3)

The Model (b) is specified as:

$$\ln(\text{pc_protein}) = \beta_0 + \beta_1 \ln(\text{hh_size}) + \beta_2 \ln(\text{mpce})$$

Table 3.2: Coefficients of Model (b)

	<i>Estimate</i>	<i>Std_Error</i>	<i>t_value</i>	<i>p_value</i>
<i>Intercept</i>	0.6649	0.1329	5.0039	6.12e-07 ***
<i>ln_hh_size</i>	-0.1168	0.0100	-11.6713	< 2e-16 ***
<i>ln_mpce</i>	0.4228	0.0148	28.6260	< 2e-16 ***

Model (b) regresses the logarithm of per capita protein consumption on the logarithm of household size and the logarithm of MPCE. The intercept is estimated at 0.6649, with a standard error of 0.1329 and a t-statistic of 5.004, indicating statistical significance at the 1 per cent level. The coefficient on $\ln(\text{household size})$ is -0.1168 , with a standard error of 0.01001 and a t-statistic of -11.671 , which is also significant at the 1 per cent level. This suggests that a 1 per cent increase in household size reduces per capita protein consumption by approximately 0.12 per cent, keeping everything else constant. The coefficient on $\ln(\text{mpce})$ is 0.4228, with a standard error of 0.01477 and a t-statistic of 28.626, and is highly significant at the 1 per cent level. Thus, a 1 per cent increase in MPCE increases protein consumption by about 0.42 per cent, keeping everything else constant. The R-square of this model is 0.4623, and the Adjusted R-square is 0.4618. Thus, Model (b) explains a larger proportion of the variation in protein consumption than Model (a).

Comparing the two models, Model (b) has a higher R-square (0.4623 versus 0.4346) and a higher Adjusted R-square (0.4618 versus 0.4341). It also has a substantially lower AIC value (-672.49 compared to -574.49). Since a lower AIC indicates better model fit, Model (b) clearly provides the superior specification. Additionally, Model (b) allows elasticity interpretation for both household size and income, which is economically more meaningful. Therefore, Model (b) is preferred.

4. In the regression model of $\ln(\text{Per capita ProteinHH})$ regressed on hh_size and $\ln_mpce$, compare the results for the model with and without intercept and comment.

The Model with intercept is specified as:

$$\ln(\text{pc_protein}) = \beta_0 + \beta_1 \text{hh_size} + \beta_2 \ln(\text{mpce})$$

Table 4.1: Coefficients of Model with Intercept

Estimate Std_Error t_value p_value

<i>Intercept</i>	0.1785	0.1305	1.3680	0.1715
<i>hh_size</i>	-0.0112	0.0019	-5.8410	6.07e-09 ***
<i>ln_mpce</i>	0.4673	0.0149	31.4591	< 2e-16 ***

In the above model, the estimated intercept is 0.1785, with a standard error of 0.1305 and a t-statistic of 1.368, indicating that it is not statistically significant. The coefficient on household size is -0.0112 , with a standard error of 0.001918 and a t-statistic of -5.841 , showing strong statistical significance. It indicates that an additional household member reduces per capita protein consumption by approximately 1.12 per cent, holding everything else constant. The coefficient on $\ln(\text{MPCE})$ is 0.4673, with a standard error of 0.01485 and a t-statistic of 31.459, which is highly significant. This implies that a 1 per cent increase in MPCE increases per capita protein consumption by approximately 0.47 per cent, keeping everything else constant. The R-square of this model is 0.4346, and the Adjusted R-square is 0.4341. The AIC value is -574.49 .

The Model without intercept is specified as:

$$\ln(\text{pc_protein}) = \beta_1 \text{hh_size} + \beta_2 \ln(\text{mpce})$$

Table 4.2: Coefficients of Model without Intercept

	<i>Estimate</i>	<i>Std_Error</i>	<i>t_value</i>	<i>p_value</i>
<i>hh_size</i>	-0.0099	0.0017	-5.9598	2.99e-09 ***
<i>ln_mpce</i>	0.4875	0.0010	503.7887	< 2e-16 ***

In the above model, the coefficient on household size is -0.009885 , with a standard error of 0.001659 and a t-statistic of -5.96 , remaining statistically significant at the 1 per cent level. A one-unit increase in household size (i.e., one additional household member) decreases per capita protein consumption by approximately 0.99 per cent. The coefficient on $\ln(\text{mpce})$ increases to 0.4875, with a standard error of 0.000968 and an extremely large t-statistic of 503.79, indicating

higher statistical significance. A 1 percent increase in MPCE leads to a 0.4875 percent increase in per capita protein consumption. The reported R-square is 0.9978; however, this value is not directly comparable to the R-square from the model with an intercept (because R computes R-square differently when the intercept is omitted). In such cases, R-squared is often artificially inflated. The AIC for the model without the intercept is -574.62 , which is almost identical to that of the model with the intercept.

Since the AIC values are nearly the same and the very high R-square in the no-intercept model is misleading, there is no strong statistical justification for removing the intercept. Moreover, excluding the intercept forces the regression line to pass through the origin, which imposes a restrictive assumption that lacks economic justification. Therefore, even though the intercept is not statistically significant, the model with the intercept is preferred because it is less restrictive and consistent with standard econometric practice.

APPENDIX**R CODES****Q1)**

Installation of Required packages

install.packages("readr")

library(readr)

install.packages("psych")

library("psych")

install.packages("readxl")

library(readxl)

Importing and viewing the dataset

setwd("C:\\Users\\Kanmani S\\Downloads")

Assignment1Data <- read_excel("Assignment1Data.xls",

range = "A1:K1954")

View(Assignment1Data)

#a)

Assignment1Data\$per_wgt <- Assignment1Data\$hh_size * Assignment1Data\$weights

Run regression

model1 <- lm(ProteinHH ~ mpce,

data = Assignment1Data,

weights = Assignment1Data\$per_wgt)

summary(model1)

```
summary(model1)$r.squared
```

```
# elasticity
```

```
mean_protein <- mean(Assignment1Data$ProteinHH, na.rm = TRUE)
```

```
mean_mpce <- mean(Assignment1Data$mpce, na.rm = TRUE)
```

```
elasticity1 <- coef(model1)[2] * (mean_mpce / mean_protein)
```

```
elasticity1
```

```
#b)
```

```
# Create per capita protein
```

```
Assignment1Data$pc_protein <- Assignment1Data$ProteinHH /
```

```
  Assignment1Data$hh_size
```

```
# Create weights
```

```
Assignment1Data$per_wgt <- Assignment1Data$hh_size *
```

```
  Assignment1Data$weights
```

```
# Run regression
```

```
model2 <- lm(pc_protein ~ mpce,
```

```
  data = Assignment1Data,
```

```
  weights = per_wgt)
```

```
summary(model2)
```

```
summary(model2)$r.squared
```

```
# elasticity
```

```
mean_pc_protein <- mean(Assignment1Data$pc_protein, na.rm = TRUE)
```

```
elasticity2 <- coef(model2)[2] * (mean_mpce / mean_pc_protein)
```

elasticity2

#c)

(log-log model)

```
Assignment1Data <- subset(Assignment1Data, pc_protein > 0 & mpce > 0)
```

```
Assignment1Data$ln_pc_protein <- log(Assignment1Data$pc_protein)
```

```
Assignment1Data$ln_mpce <- log(Assignment1Data$mpce)
```

Run regression

```
model3 <- lm(ln_pc_protein ~ ln_mpce,
```

```
  data = Assignment1Data,
```

```
  weights = per_wgt)
```

```
summary(model3)
```

```
summary(model3)$r.squared
```

#d)

(log-linear model)

Run regression

```
model4 <- lm(ln_pc_protein ~ hh_size,
```

```
  data = Assignment1Data,
```

```
  weights = per_wgt)
```

```
summary(model4)
```

```
summary(model4)$r.squared
```

elasticity


```

coef(model4)["hh_size"]

weighted.mean(Assignment1Data$hh_size, Assignment1Data$per_wgt)

mean_hh <- weighted.mean(Assignment1Data$hh_size, Assignment1Data$per_wgt)

elasticity <- coef(model4)["hh_size"] *

  weighted.mean(Assignment1Data$hh_size,

    Assignment1Data$per_wgt)

elasticity

```

#e)

(Multiple regression model)

Run regression

```

model5 <- lm(ln_pc_protein ~ hh_size + ln_mpce,

  data = Assignment1Data,

  weights = per_wgt)

summary(model5)

summary(model5)$adj.r.squared

```

Q2)

a)

Loading of required libraries

```

library(haven)

asgn1_cal_prot <- read_dta("C:/Users/Gugan/Downloads/asgn1-cal-prot.dta")

```

```
View(asgn1_cal_prot)

library(haven)

library(dplyr)

library(fixest)

library(modelsummary)


# Import and view dataset

data <- read_dta("asgn1-cal-prot.dta")

library(haven)

data <- read_dta("asgn1-cal-prot.dta")

# Inspect data structure and variables

names(data)

unique(data$state)

class(data$state)

table(data$state)

names(data)


# Filter data for Andhra Pradesh (State=28)

ap <- data %>% filter(state == 28)


# Create required variables

ap <- ap %>%

  mutate(
```

```

per_wgt = hh_size * weights,
pc_protein = ProteinHH / hh_size,
pc_energy = EnergyHH / hh_size,
pc_fat = FatHH / hh_size,
ln_pc_protein = log(pc_protein),
ln_pc_energy = log(pc_energy),
ln_pc_fat = log(pc_fat),
ln_mpce = log(mpce)
)

# Estimation of regression model for protein,energy and fat
library(fixest)

m_protein <- feols(ln_pc_protein ~ hh_size + ln_mpce,
  data = ap, weights = ~per_wgt)

m_energy <- feols(ln_pc_energy ~ hh_size + ln_mpce,
  data = ap, weights = ~per_wgt)

m_fat <- feols(ln_pc_fat ~ hh_size + ln_mpce,
  data = ap, weights = ~per_wgt)

# Display regression results

```

```
library(modelsummary)
```

```
modelsummary(
  list("Protein" = m_protein,
       "Energy" = m_energy,
       "Fat" = m_fat),
  stars = TRUE
)
```

Q3)

```
install.packages("readr")
```

```
library(readr)
```

```
install.packages("psych")
```

```
library("psych")
```

```
install.packages("readxl")
```

```
library(readxl)
```

```
setwd("C:\\Users\\dell\\Downloads")
```

```
Assignment1Data <- read_excel("Assignment1Data.xls",
                              range = "A1:K1954")
```

```
View(Assignment1Data)
```

```
head(Assignment1Data$ProteinHH)
```

```
head(Assignment1Data$pc_protein)
```

```
head(Assignment1Data$hh_size)
```

```

min(Assignment1Data$hh_size)

#Create required variables

Assignment1Data$pc_protein <- Assignment1Data$ProteinHH / Assignment1Data$hh_size
Assignment1Data$ln_pc_protein <- log(Assignment1Data$pc_protein)
Assignment1Data$ln_mpce <- log(Assignment1Data$mpce)
Assignment1Data$ln_hh_size <- log(Assignment1Data$hh_size)
Assignment1Data$per_wgt <- Assignment1Data$hh_size * Assignment1Data$weights
head(Assignment1Data$per_wgt)
summary(Assignment1Data$per_wgt)
names(Assignment1Data)

#Estimate Model (a)

model_a <- lm(ln_pc_protein ~ hh_size + ln_mpce,
              data = Assignment1Data,
              weights = per_wgt)
summary(model_a)

# Tabulate results of Model (a)

summary_model_a <- summary(model_a)
coef_table <- cbind(
  Estimate = summary_model_a$coefficients[,1],
  Std_Error = summary_model_a$coefficients[,2],

```

```

t_value = summary_model_a$coefficients[,3],
p_value = summary_model_a$coefficients[,4]
)
print(round(coef_table, 4))

```

```

#Estimate Model (b)

```

```

model_b <- lm(ln_pc_protein ~ ln_hh_size + ln_mpce,
              data = Assignment1Data,
              weights = per_wgt)
summary(model_b)

```

```

# Tabulate results of Model (b)

```

```

summary_model_b <- summary(model_b)
coef_table <- cbind(
  Estimate = summary_model_b$coefficients[,1],
  Std_Error = summary_model_b$coefficients[,2],
  t_value = summary_model_b$coefficients[,3],
  p_value = summary_model_b$coefficients[,4]
)
print(round(coef_table, 4))

```

```

#Compare Models (a) and (b)

```

```

AIC(model_a, model_b)

```

Q4)

#Estimate Model with intercept

```
model_with_intercept <- lm(ln_pc_protein ~ hh_size + ln_mpce,
  data = Assignment1Data,
  weights = per_wgt)
summary(model_with_intercept)
```

#Estimate Model without intercept

```
model_no_intercept <- lm(ln_pc_protein ~ hh_size + ln_mpce - 1,
  data = Assignment1Data,
  weights = per_wgt)
summary(model_no_intercept)
```

Tabulate results of Model with no intercept

```
summary_model_no_intercept <- summary(model_no_intercept)
coef_table <- cbind(
  Estimate = summary_model_no_intercept$coefficients[,1],
  Std_Error = summary_model_no_intercept$coefficients[,2],
  t_value = summary_model_no_intercept$coefficients[,3],
  p_value = summary_model_no_intercept$coefficients[,4]
)
```

```
print(round(coef_table, 4))
```

```
#Comparing Models with and without intercept
```

```
AIC(model_with_intercept, model_no_intercept)
```